Krishiv Patel

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Education

Rutgers University, School of Engineering Piscataway, NJ
B.S. Aerospace Engineering, Minor in Mathematics, Energy Systems Certificate Expected May 2027

- GPA 3.44/4.00 | Dean's List: 4 semesters
- James J. Slade Scholar: graduate research program, 6 credits of graduate independent research
- Honors: Francisco Diaz-Masso Engineering Scholarship, NASA Space Grant Scholarship, Rutgers School of Engineering Scholarship

Experience

Rutgers Department of Mechanical & Aerospace Engineering Piscataway, NJ
NASA Space Grant Fellow (2026–27) | Research Lead, Design & CFD (from Undergraduate Researcher) Mar 2025 – Present

- Lead design on a $45K pulsed-air actuator replacing fixed nacelle chines with deployable active flow control: 150–200 m/s jets at 10–1000 Hz, deployed only for takeoff and landing to avoid the chine's cruise drag
- Re-architected after a manufacturability review with Calspan and GE engineers: a localized plenum replaced the external compressor feed, decoupling pressure from modulation; aerodynamics reviewed by two former Boeing engineers including a Technical Fellow
- Own CFD (ANSYS Fluent, CFX) and parametric CAD (SolidWorks); tolerance to ASME Y14.5-2018, backed by Rutgers Makerspace certification on CNC and laser cutting

Rutgers Rocket Propulsion Lab Piscataway, NJ
Design & Aerodynamics, 750 lbf Lab Engine Program Mar 2025 – Present

- Size coolant channels and run flow analysis for regenerative cooling on the lab's 750 lbf pressure-fed N2O/IPA engine, which hot fired in May 2026; resulting design changes cut chamber wall temperature 3%
- Size injector orifices by hand calculation to keep regenerative coolant passages thermally isolated from the solid motor igniter

NASA New Jersey Space Grant Consortium – Rutgers University Piscataway, NJ
Experimental Aerodynamics Research Intern May 2026 – Aug 2026

- Directed five Governor's School of Engineering & Technology scholars, from the top 64 high school engineers in New Jersey, on a NASA-funded study open only to James J. Slade Scholars
- Built the test setup, calibration procedure, and data reduction workflow; ran CAD, CFD, 3D printing, and wind tunnel testing across 51 biplane configurations in two test matrices at Re = 20,100, with finite-span corrections
- Peak L/D 9.12 across four trials vs 4.21 monoplane; corresponding author on both resulting papers

Rutgers Department of Mechanical & Aerospace Engineering Piscataway, NJ
Wind Tunnel Lab Developer Aug 2026 – Present

- Co-developing the wind tunnel module for the Rutgers MAE Senior Lab, a required course for all Aerospace and Mechanical seniors
- Redesigned and reprinted the airfoil model set for repeated use; documented the test's limitations so students can run it unsupervised

Publications

Experimental Investigation of Airfoil Aerodynamics and Flow Control Concepts: Mono and Biplane Configurations

- IEEE MIT Undergraduate Research Technology Conference, presented Oct 2026 | first author; co-corresponding with Dr. DeMauro
- GSET Research Journal, published July 2026 | sole corresponding author

Projects

750 lbf IPA/N2O Bipropellant Rocket Engine *(Head of Design, 8-person team, Rutgers Senior Design)* May 2026 – Present

- Own injector plate design: converted element geometry from coaxial swirl to shear coaxial for machinability and uniform propellant distribution, and redesigned the igniter interface as a replaceable canister so the team can iterate between hot fires
- Diagnosed prior-generation engine: 50 lbf against a 750 lbf target, traced to unsustained chamber ignition driving combustion downstream of the throat with no choked flow; redesigning igniter integration for sustained in-chamber combustion
- Own vehicle integration and the static test stand for cold flow and hot fire, including feed system design and high-pressure procedures

Spacecraft Thermal Subsystem *(NASA L'SPACE Mission Concept Academy)* Aug 2026 – Present

- Own the thermal subsystem for a robotic lunar south pole payload exploring permanently shadowed regions near 40 K: thermal control architecture, environment definition, and component temperature margins
- Designing to a 250 kg, $250M envelope with no RTG power, through gate reviews to a NASA-reviewed Preliminary Design Review

Hypersonic Nose Cone Aerothermal Analysis *(University Consortium for Applied Hypersonics)* Jul 2025 – Dec 2025

- Analyzed multiple nose cone geometries at hypersonic conditions to characterize shockwave structure and aerodynamic heating, splitting flow solution in ANSYS Fluent with conduction analysis in ANSYS Mechanical (Transient Thermal)
- Ran a blunt vs sharp trade of drag against stagnation heating; integrated an internal honeycomb for conductive heat dissipation

Technical Skills

Analysis: ANSYS Fluent, ANSYS CFX, ANSYS Mechanical (Transient Thermal), XFLR5, MATLAB, Python, LabVIEW; classical lamination theory, composite failure criteria (Hashin, max stress)
Design: SolidWorks (CSWA), OnShape, ANSYS SpaceClaim, GD&T (ASME Y14.5-2018), Design for Manufacturability
Experimental & Fabrication: Wind tunnel testing and calibration, hot-wire anemometry, smoke flow visualization, Shopbot CNC, laser cutting, FDM 3D printing